

Engine Will Not Stop From the Helm or Customer Interface Box

Symptoms

- Engine will **not** shutoff when the Stop or Stop/Start is switched to OFF at the helm or customer interface box (CIB).

How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Possible Causes:

- Malfunctioning Stop switch
- Open circuit in the Stop signal
- Short circuit in the Stop signal.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the helm wiring harness.	
	STEP 1A. Check the START/STOP switch/button (helm).	Less than 10 ohms resistance when switch is in ON/Pushed position?
	STEP 1B. Inspect the helm harness.	Dirty or damaged pins?
	STEP 1C. Check the helm wiring harness for a pin-to-pin short in the keyswitch.	Greater than 100k ohms resistance?
	STEP 1D. Check the helm wiring harness for an open in the Stop signal.	Less than 10 ohms resistance?
STEP 2.	Check main extension wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Inspect the main extension harness.	Dirty or damaged pins?
	STEP 2B. Check the main extension wiring harness for a pin-to-pin short in the keyswitch.	Greater than 100k ohms resistance?
	STEP 2C. Check the main extension wiring harness for an open in the Stop signal.	Less than 10 ohms resistance?
STEP 3.	Check the C.I.B.	
	STEP 3A. Check the Start/Stop switch (C.I.B).	Less than 10 ohms resistance when switch is in ON position?
	STEP 3B. Check the Start/Stop switch (C.I.B).	Only shutdown relay ON?

STEP 1. Check the helm wiring harness.

STEP 1A. Check the START/STOP switch/button (helm).

Conditions :

- Disconnect the START/STOP switch/button at the helm.

Action	Specification/Repair	Next Step
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Measure the resistance. Measure the resistance between pin 3 and pin 4 at the Stop switch pigtail harness. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms resistance when switch is in ON/Pushed position? YES	1B
	Less than 10 ohms resistance when switch is in ON/Pushed position? NO Repair : Replace the start switch. For PS102 Systems:. Refer to Procedure 015-101 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-101.html) For PS103 Systems:. Refer to Procedure 015-109 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-109.html)	Repair complete.

STEP 1B. Inspect the helm harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect the helm harness from the main extension harness.

Action	Specification/Repair	Next Step
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• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • For the helm harness: Refer to Procedure 015-078 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-078.html) • For the main extension harness: Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete
	Dirty or damaged pins? NO	1C

STEP 1C. Check the helm wiring harness for a pin-to-pin short in the keyswitch.

Conditions :

- Turn system enable switch OFF.
- Disconnect all mating harnesses to the helm harness.

Action	Specification/Repair	Next Step
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Measure the resistance between the Stop SIGNAL pin 3 in the helm harness connector (mating to the main extension) and all other pins in the helm harness connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms resistance? YES	1D
	Greater than 100k ohms resistance? NO Repair : A short circuit has been detected in the keyswitch. Repair or replace helm wiring harness. Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete

STEP 1D. Check the helm wiring harness for an open in the Stop signal.

Conditions :

- Turn system enable switch OFF.
- Disconnect all mating harnesses to the helm harness.

Action	Specification/Repair	Next Step

Measure the resistance between the Stop SIGNAL pin 9 at the helm harness connector (mating to the main extension harness) and Stop SIGNAL pin 3 at the helm harness connector (mating to the Stop switch). Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms resistance? YES	2A
	Less than 10 ohms resistance? NO Repair : A open circuit has been detected in the keyswitch. Repair or replace helm wiring harness. Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete

STEP 2. Check main extension wiring harness.

STEP 2A. Inspect the main extension harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect the main extension harness from the engine interface harness.

Action	Specification/Repair	Next Step

• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. • For the main extension harness: Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete
	Dirty or damaged pins? NO	2B

STEP 2B. Check the main extension wiring harness for a pin-to-pin short in the keyswitch.

Conditions : • Turn system enable switch OFF. • Disconnect all mating harnesses to the main extension harness.		
Action	Specification/Repair	Next Step

Measure the resistance between the Stop SIGNAL pin 9 in the main extension harness connector and all other pins in the main extension harness connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms resistance? YES	2C
	Greater than 100k ohms resistance? NO Repair : A short circuit has been detected in the Stop signal. Repair or replace main extension wiring harness. Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete.

STEP 2C. Check the main extension wiring harness for an open in the Stop signal.

Conditions :

- Turn system enable switch OFF.
- Disconnect the C.I.B. from the main extension wiring harness.
- Disconnect the helm harness from the main extension wiring harness

Action	Specification/Repair	Next Step
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Measure the resistance between the Stop SIGNAL pin 9 at the main extension wiring harness connector (mating to the C.I.B) and Stop SIGNAL pin 9 at the main extension wiring harness connector (mating to the helm harness). Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : An open circuit has been detected in the Stop signal. Repair or replace the main extension wiring harness. Refer to Procedure 015-077 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-077.html)	Repair complete.

STEP 3. Check the C.I.B.

STEP 3A. Check the START/STOP switch (C.I.B).

Conditions :

- Open up the customer interface Box. Refer to Procedure 015-023 in Section 15.
(/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)
- Disconnect the START/STOP switch. Refer to Procedure 015-109 in Section 15.
(/qs3/pubsys2/xml/en/procedures/513/513-015-109.html)

Action	Specification/Repair	Next Step

Measure the resistance between pin 2 and pin 3 at the START/STOP switch connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms resistance when switch is in ON position? YES	3B
	Less than 10 ohms resistance when switch is in ON position? NO Repair : Replace the START/STOP switch. Refer to Procedure 015-109 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-109.html)	Repair complete

STEP 3B. Check the START/STOP switch (C.I.B.).

Conditions :

- Open up the C.I.B. Refer to Procedure 015-023 in Section 15.
(/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)
- Press and hold START/STOP button.

Action	Specification/Repair	Next Step
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Monitor shutdown relay LED.	Only shutdown relay ON? YES Repair : Refer to the Engine Will Not Shutoff in the troubleshooting symptom tree in Section TS	Repair complete
	Only shutdown relay ON? NO Repair : A malfunction has been detected in the C.I.B. Replace the C.I.B. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	Repair complete